

```
import pandas as pd
```

```
df = pd.read_csv("spam.csv")
df.head()
```

	Category	Message	
0	ham	Go until jurong point, crazy.. Available only ...	
1	ham	Ok lar... Joking wif u oni...	
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	
3	ham	U dun say so early hor... U c already then say...	
4	ham	Nah I don't think he goes to usf, he lives aro...	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.groupby('Category').describe()
```

	Message			
	count	unique	top	freq
Category				
ham	4825	4516	Sorry, I'll call later	30
spam	747	641	Please call our customer service representativ...	4

```
df['spam']=df['Category'].apply(lambda x: 1 if x=='spam' else 0)
df.head()
```

	Category	Message	spam	
0	ham	Go until jurong point, crazy.. Available only ...	0	
1	ham	Ok lar... Joking wif u oni...	0	
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	1	
3	ham	U dun say so early hor... U c already then say...	0	
4	ham	Nah I don't think he goes to usf, he lives aro...	0	

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```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(df.Message,df.spam)
```

```
from sklearn.feature_extraction.text import CountVectorizer
v = CountVectorizer()
X_train_count = v.fit_transform(X_train.values)
X_train_count.toarray()[ :2]
```

```
array([[0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0]])
```

```
from sklearn.naive_bayes import MultinomialNB
model = MultinomialNB()
model.fit(X_train_count,y_train)
```

```
▼ MultinomialNB ⓘ ?
MultinomialNB()
```

```
emails = [
    'Hey mohan, can we get together to watch footbal game tomorrow?',
    'Upto 20% discount on parking, exclusive offer just for you. Dont miss this reward!'
]
emails_count = v.transform(emails)
model.predict(emails_count)
```

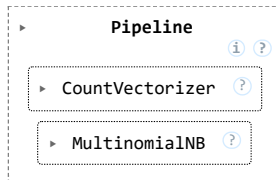
```
array([0, 1])
```

```
X_test_count = v.transform(X_test)
model.score(X_test_count, y_test)
```

```
0.990667623833453
```

```
from sklearn.pipeline import Pipeline
clf = Pipeline([
    ('vectorizer', CountVectorizer()),
    ('nb', MultinomialNB())
])
```

```
clf.fit(X_train, y_train)
```



```
clf.score(X_test,y_test)
```

```
0.990667623833453
```

```
clf.predict(emails)
```

```
array([0, 1])
```